MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025 THIRD SEMESTER

Discipline Specific Core Course (DSE)-MG3DSEECO201- FINANCIAL ECONOMICS (2024 ADMISSION ONWARDS) SCHEME OF VALUATION

Duration: 2 Hours Maximum Marks:70

Part A(Answer all questions,1 mark each)

1. B) Future
2. C) Present value of benefits / Present value of costs
3. B) One party has more or better information than the other
4. C) Moral hazard
5. B) Compounding
6. A) One period
7. C) Investors expect short-term interest rates to fall
8. D) All the above
9. B) An average of current and expected future short-term interest rates
10. c) Between investors

Part B(Answer any 10 questions,3 mark each)

11. Risk diversification is a strategy that involves investing in a variety of financial assets, so you can reduce your exposure to any one specific type of risk. The basic principle behind diversification is that by including a mix of asset types in your portfolio, rather than concentrating on just one, you can reduce the negative effects of market volatility or poor performance by one specific asset.

12. The term structure of interest rates, which is the relationship between interest rates and different maturities, directly forms the yield curve.. The shape of the yield curve—upward-sloping, downward-sloping (inverted), or flat—is a direct visual representation of this term

Structure and reflects market expectations for future interest rate.

OR

A normal upward-sloping curve indicates expectations of economic growth and potentially higher future rates, while an inverted curve suggests the opposite, often signalling an upcoming recession.

13. Systematic risk is market-wide, inherent to the overall market, and cannot be eliminated through diversification eg: market risk, interest rate risk etc. While unsystematic risk is specific to a company or industry and can be reduced or eliminated through diversification. Eg: business risk , financial risk etc **OR**

Systematic Risk

Non-diversifiable / Market / Macroeconomic risk,Caused by economy-wide factors,Macro in nature,Cannot be reduced by diversification,Measured by Beta,Includes market risk, interest rate risk, purchasing power risk etc

Unsystematic Risk

Diversifiable / Specific / Microeconomic risk,Caused by firm/industry-specific factors,Micro in nature,Can be reduced through diversification,No standard measure like Beta,Includes business risk and financial risk etc **(Any 3 differences)**

14. Carrying costs in inventory management are generally classified into four main categories:

1. Capital Costs

Interest, opportunity cost etc

2. Storage Space Costs

Warehousing,record keeping,
etc

3. Inventory

Service Taxes,
insurance

4. Inventory Risk Costs

Absolete ,damaged, stolen or unsellable

15. Liquidity management is the strategic planning and control of an organization's liquid assets and financial resources to ensure its ability to meet short-term obligations and capitalize on potential opportunities. It is a crucial component of financial management. It involves monitoring, forecasting, and optimizing the availability of liquid assets, like cash and marketable securities.

16. **Indicator of Market Performance**: Stock indices like NIFTY or SENSEX show overall market movements. Investors use them to understand whether the market is rising (bullish) or falling (bearish).

Benchmark for Portfolio Evaluation:Investors compare the return on their investments with index returns to assess how well their portfolio is performing.

Basis for Investment Decisions: Indices help identify trends in specific sectors or the whole economy, guiding investors on when to buy, hold, or sell securities.

17 .A business with a residual dividend policy holds zero excess cash at any given point in time. All spare cash must be either reinvested in the business or redistributed among the shareholders. Imperfections in the capital market make it rare for a company to follow a pure residual dividend policy. Most businesses instead follow smooth dividend policies that call for regular dividends that show some correlation with the business' past and present earnings.

18.Goal setting refers to the process of identifying an individual's financial objectives and planning strategies to achieve them. It helps in determining how much money is needed, by when, and through which financial instruments. Financial goals may be short-term (e.g., emergency funds), medium-term (e.g., buying a vehicle), or long-term (e.g., retirement planning).A clear financial goal guides saving and investment decisions, ensures better resource allocation, reduces unnecessary spending, and helps track progress. Effective goal setting follows the SMART principle — specific, measurable, achievable, realistic, and time- bound.(**Any other relevant points)**

19. The One Fund Theorem states that all investors will hold the same optimal risky portfolio irrespective of their individual risk preferences. In other words, among all risky assets in the market, there exists one unique efficient risky portfolio that every investor should invest in.

Individual investors then combine this optimal risky portfolio with the risk-free asset depending on their risk tolerance. More risk-loving investors will invest more in the risky portfolio, while risk-averse investors will allocate more to the risk-free asset. This theorem

simplifies portfolio selection by suggesting that security analysis should mainly focus on identifying this single best risky portfolio, rather than building separate portfolios for each investor.(Any other relevant points)

20.The risk-free asset plays a crucial role in the Capital Market Line. It allows investors to combine a risk-free return with a market portfolio. By investing partly in the risk-free asset and partly in the market portfolio, investors can achieve any desired risk–return combination on the CML. Low-risk investors lend (invest) in the risk-free asset, while high-risk investors borrow at the risk-free rate to invest more in the market portfolio.Thus, the risk-free asset helps create a straight-line relationship between risk and return, improving portfolio efficiency and giving all investors access to an optimal portfolio on the CML.

21.Price Determination-Funds Mobilization- Liquidity-Risk sharing-Early Acess-Reduction in transaction costs and provision of the Information-Capital Formation(**Any other relevant points,any 3 or 4 functions)**

22. The Modigliani-Miller theorem (MsM) was developed in the 1650s by Merton Miller and Franco Modigliani. This important and still relevant theory states that the market value of a company is calculated as the present value of its future earnings and underlying assets. They believed that taxes and government regulations do not contribute to capital structure, and that market value is independent of a company's capital structure. In 1663, Modigliani and Miller revisited their theorem by incorporating corporate taxes. This adjustment fundamentally changes the conclusion: the value of a firm is no longer independent of its capital structure. Instead, debt financing becomes advantageous due to the tax deductibility of interest payments.

23. Derivatives originated as tools for managing price risks in commodities like agricultural products. The earliest forms of derivative trading were forward contracts, which existed centuries ago for crops and precious metals. Organized derivative markets developed in the 16th century, notably with the establishment of the Chicago Board of Trade (CBOT) in 1848.Later, standardized futures contracts and regulated exchanges expanded derivative trading. In the late 20th century, financial derivatives such as options, swaps, and index futures emerged, driven by financial market globalization, technological advancements, and the need for sophisticated risk management.(VALUE LIBERALLY)

24. Market Risk Premium refers to the additional return that investors expect to earn from investing in the market portfolio instead of a risk-free asset. It compensates investors for taking higher market risk.It is calculated as: Market Risk Premium=Expected Return on Market-Risk-Free Rate (eqn is not compulsory) It is a key component in CAPM and helps determine the required rate of return on risky assets.

25. Adverse selection occurs before a transaction takes place. It arises due to hidden information, meaning one party has better knowledge about their characteristics than the other. Because the buyer or seller cannot observe this private information in advance, the party with higher risk is more likely to participate.Example: In insurance, unhealthy individuals are more likely to buy health insurance than healthy ones.(Any other relevant points)

PART C(Answer any 5 questions, 6 marks each)

26. The Two-Fund Theorem (Mutual Fund Theorem) states that any efficient portfolio can be constructed as a linear combination of any two efficient portfolios, usually a risk-free asset and a market portfolio. This simplifies portfolio construction significantly.

1. Reduces complexity in portfolio selection-Investors do not need to analyze and choose from thousands of individual securities. They only need two funds:

- One risk-free fund
- One optimal risky fund (market portfolio)Thus, portfolio construction becomes easy and practical.

2. Separates investment and financing decisions-The investment decision determines the optimal risky portfolio based on asset returns, risks, and correlations.

3. Suitable for all types of investors-Risk-averse investors lend (invest) more in the risk-free asset.Risk-tolerant investors borrow at the risk-free rate and invest more in the market portfolio.So, one optimal risky portfolio works for everyone, with only the allocation changing.

4. Forms the basis of the Capital Market Line (CML)-By combining a risk-free asset and market portfolio, we get a straight line — the CML — showing the best risk-return combinations available.(Give marks to any other relevant points also)

27. Cost of equity is the minimum rate of return required by shareholders to compensate for the risk of investing in a company's equity. Methods to calculate cost of equity include the Capital Asset Pricing Model (CAPM) and the Dividend Discount Model (DDM). Students face this concept in ratio analysis and case-based exams. Formula for Cost of Equity Cost of Equity = Risk-free Rate + Beta × (Market Return – Risk-free Rate) **Cost of capital** is the average rate a firm pays for financing from all capital sources— both equity and debt. It is usually calculated

using the Weighted Average Cost of Capital (WACC) formula and is central to deciding if a project or investment makes financial sense. Weighted Average Cost of Capital (WACC) Formula WACC = (E/V) × Cost of Equity + (D/V) × Cost of Debt × (1 – Tax Rate)

E = Market value of equity ,D = Market value of debt V = E + D (total capital)

28.Asset–Liability Management (ALM)-Matching maturity and repricing of assets and liabilities to maintain stability in net interest income.

Hedging Strategies-Using derivatives like interest rate swaps, futures, options, and forwards to lock in or protect against rate changes.

Diversification of Assets s Liabilities-Holding a mix of fixed-rate and floating-rate instruments to spread risk across different interest-sensitive items.

Duration Matching-Matching the duration of assets and liabilities to minimize the effect of interest rate changes on portfolio value.

Gap Management-Monitoring interest rate sensitive assets and liabilities gaps (RSA

– RSL) and adjusting to reduce exposure.

Securitization-Converting long-term loans into marketable securities to transfer risk off the balance sheet.

Repricing Strategies-Frequently revising interest rates on loans and deposits to align with market interest rates.

29.Futures and forward contracts are both derivative instruments used for hedging and speculation, but they differ in key aspects. A futures contract is a standardized agreement traded on an organized exchange such as the NSE or CME, where the quantity, quality, and delivery date of the underlying asset are predefined. On the other hand, a forward contract is a customised, private agreement between two parties, usually traded in the over-the- counter (OTC) market, allowing terms to be tailored to the needs of the buyer and seller.

In futures contracts, the exchange acts as an intermediary and ensures performance through a clearing house, which reduces counterparty risk. Forward contracts involve high

credit risk, as there is no intermediary and the settlement depends solely on the parties involved. Futures contracts require daily settlement, known as mark-to-market, where gains or losses are

settled each day, while forward contracts are settled only at maturity. Futures are highly liquid due to active secondary trading on exchanges, whereas forwards are less liquid because they are private agreements and difficult to transfer.

Futures prices are transparent and regulated, while forward prices remain confidential. Overall, futures are suitable for traders requiring liquidity and risk reduction, while forwards benefit parties wanting customisation and flexibility.

30.Protective Functions

These functions help protect investors from cheating and fraud.

a. Bans Insider Trading-Insider trading means using secret infomation to make profits in the market. SEBI bans this. It makes sure everyone gets equal and fair information.

b. Stops Fraud and Unfair Practices-SEBI checks for illegal actions like price rigging. It takes strong steps against such activities. It helps keep the market honest.

c. Educates Investors-SEBI runs programs to educate investors. It helps people understand their rights. It teaches how to make smart investments.

d. Protects Small Investors-SEBI focuses on protecting small investors. It ensures that companies do not cheat or mislead them.

2. Regulatory Functions

These functions help SEBI control and manage the market system.

a. Registration of Market Players-SEBI registers all players in the market. These include brokers, dealers, mutual funds and stock exchanges.

b. Frames Rules and Guidelines-SEBI makes rules for how the market works. All companies and brokers must follow these rules.

c. Checks Company Activities-SEBI monitors mergers, takeovers and public issues. It makes sure that no company takes advantage of investors.

d. Audits and Inspections-SEBI conducts regular inspections. It checks the records of market participants. If someone breaks rules, SEBI can punish them.

3. Developmental Functions

These functions help improve the market and its systems.

a. Promotes Use of Technology-SEBI supports online trading and use of computers. It helps improve the speed and safety of trading.

b. Encourages New Products-SEBI allows companies to introduce new investment products. This gives more options to investors.

c. Supports Self-Regulation-SEBI allows some organisations to make their own rules. These are known as Self-Regulatory Organisations (SROs). SEBI watches over them.

d. Organises Training-SEBI arranges training programs for brokers and investors. It helps them learn new skills and stay updated(any other relevant points)

31. Risk management is the systematic process of identifying, analyzing, and controlling risks that may adversely affect the financial performance or objectives of an organization. In the modern business environment, firms are exposed to various uncertainties such as market risks, credit risks, operational risks, and liquidity risks. Therefore, risk management aims to reduce the impact of these uncertainties and ensure the stability and sustainability of the firm.

The first step in risk management is risk identification, where all potential internal and external risks are recognized. After identifying them, the next step is risk assessment which involves measuring the probability and impact of risk using tools like Value at Risk (VaR), scenario analysis, and stress testing. Once the risk is evaluated, firms adopt risk control or mitigation strategies such as diversification, hedging, insurance, and asset-liability management to reduce exposure.

Risk management is a continuous process, so firms must regularly monitor risks and update strategies to match changing market conditions. Effective risk reporting and strong governance also help in promoting a sound risk culture within the organization.

In conclusion, risk management plays a crucial role in protecting a firm from financial instability, minimizing losses, enhancing investor confidence, and ensuring long-term growth and profitability.

32. Advantages of Bonds

1. Fixed Income: This investment instrument is characterised by its income predictability and regular interest payments, known as coupon payments. This quality pulls in investors who want a source of income with a low risk of losing money, such as retirees or individuals who don't want to take a lot of risk. By investing in the bonds, you can plan your financial obligations and cash flows much more accurately, as the payments made on interest are predetermined. They are not exposed to stock volatility in the market.

2. Capital Preservation: Bonds tend to be less fluctuant than stocks and, therefore, more attractive to investors because of their perceived ability to secure their capital, particularly for risk-averse or short-term investors. In comparison, bonds generally display fewer price fluctuations than stocks, which can be a factor as they fluctuate significantly based on market conditions and the company's performance, making them a more stable investment choice.

3. Diversification: Besides an investment portfolio, bonds can balance your investment allocation and lower the overall risk. There are certain situations when bonds behave contrary to stocks because their value is determined by, among other factors, the price of different securities and the quality of issuers' credit. Bonds offer a convenient way to smooth out the highs and lows of risk and return in a well-balanced portfolio of stocks and other real estate properties.

4. Tax Advantages: In certain situations, the types of bonds used, e. g. municipal bonds, provide tax benefits. The income earned from municipal bonds is usually free from federal income tax and local and state taxes. The unique tax-break feature can satisfy investors from higher tax brackets as it will cause more after-tax returns.

Disadvantages of Bonds

1. Interest Rate Risk: The risk related to the interest rate is one of the significant risks related to bonds. When rates go up, the value of the existing bonds with lower interest rates declines because newly issued bonds with higher rates (that look more attractive to investors) are introduced to the market. Such a risk is of particular concern to long-term bonds since they are closely related to changes in interest rates. During the sharp move in interest rates, you will sell your bonds before maturity, and this can result in selling losses.

2. Inflation Risk: Bonds generally experience the perils of inflation as well. If inflation starts to outpace the interest you earn, your investment’s real value will decrease over time. This implies that the amount you will be required to pay in interest and principal shortly will have been reduced, reducing the amount of interest you will get.

3. Reinvestment Risk: When bonds reach maturity or are redeemed by the issuer before their stated date, the reinvestment of the proceeds could be done at a lower interest rate and, hence, reduce the income streams in the future. There is more danger that it will be felt within the scenario of a decline of the interest rate, which results in lower yields than were earned by the bond.

4. Liquidity Risk: Liquidity risk means the loss that may occur when selling bonds swiftly without the danger of a substantial decrease in the bond’s price. Specific bonds, particularly those carried by smaller entities or with longer maturities, may have low liquidity. Finding buyers agreeable to giving fair value in the transaction can get tricky. It does, however, come at the risk of a lack of cash, which can be a problem when you may have to draw your money before the bond is matured.

33. Planning for retirement is an essential part of financial planning. As life expectancy increases, the need for retirement benefits increases. Retirement benefits not only secure additional income, but also help you cope with medical emergencies, meet your life aspirations, and become financially independent. Retirement is a period of life that continues to lengthen in the developed world, and the number of retirees is increasing accordingly, challenging the

strength of social security systems and social safety nets in general. Financial Retirement Planning consists of a series of activities involved in accumulating wealth to meet post-retirement needs. The negative short-term, medium-term and long-term consequences of insufficient financial planning for retirement affect not only individuals, but also their extended families and homes, which ultimately has an adverse impact on society as a whole. What is retirement planning? Retirement planning means preparing for a steady flow of money after retirement. It means setting aside funds and investing specifically with that goal in mind. Your retirement strategy will depend on your ultimate goal, income and your age. Need of retirement planning? Growing old can be expensive. Medical bills are likely to rise. Add to that the burden of inflation and not having enough money to sustain future expenses can cause stress and worry. The purpose of a retirement investment plan is to ensure financial stability in later years without depending on others. Importance if financial planning Planning for retirement does not mean that one should only focus on one's finances. Planning for retirement requires a combination of financial and personal planning. Personal planning determines a person's satisfaction in retirement.(Give marks to any oter relevant points also)

34. Yes, principal–agent problems are inevitable in large organizations because the owners (principals) and managers (agents) often have different objectives and unequal access to information. The principals expect managers to work for shareholder wealth maximization, but managers may pursue personal goals such as higher salaries, job security, or perks, leading to conflicts of interest. This occurs mainly due to information asymmetry, lack of proper monitoring, and separation of ownership and control. For example, a manager may invest in unnecessary luxury office facilities or take less risky projects only to protect their position, which may reduce shareholder returns. Similarly, in insurance companies, customers may hide their true health conditions to gain benefits, creating adverse selection.

To reduce these issues, organizations use a combination of **monitoring mechanisms and incentive alignment**. Performance-based compensation, stock options, and profit-sharing encourage managers to act in the best interests of shareholders. Corporate governance practices, independent boards, audits, and transparency in reporting also help minimize agency conflicts. In conclusion, while principal–agent problems cannot be completely eliminated in large firms, they can be effectively managed through proper incentives, accountability systems, and strong governance structures.(**Any Other relevant points)**

35. Measures systematic risk of an asset without leverage.

• Shows how much the asset's returns move relative to the overall market.

• Used in capital budgeting to assess project risk independent of financing.

• Higher asset beta → more volatile and sensitive to market fluctuations.

• Lower asset beta → more stable and less correlated with market movements.

• Helps determine appropriate discount rate in valuation decisions.

Interpretation of Beta (β)

$\beta = 1$ → Same risk as market (average sensitivity).

$\beta > 1$ → More risky than market (high sensitivity).

$\beta < 1$ → Less risky than market (low sensitivity).

$\beta = 0$ → No relation to market movements..(Any Other relevant points)